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The Estimate Lives in the Keys: What a Transformer Learns When It Adapts to Unknown Physics In-Context

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Abstract

Sequence models that adapt to unknown dynamics from their own history are widely used in robot learning, but what they compute is unclear. We study this question in small transformers trained from scratch, in a pre-registered design with exact optimal references and causal interventions, in three settings: linear systems with unknown parameters, MuJoCo pendulum, sliding-block and bouncing-ball physics, and in-context adaptive control. We prove an exact Bayes-risk identity for in-context prediction under any non-anticipating policy, which gives a parameter-free context-length law. We also prove a closed-loop identity: the information gained along unexcited directions equals the excitation energy. The trained predictors are Bayes-optimal while data are scarce, then settle at an approximation floor. Our pre-registered mechanism, a linearly readable and causally used parameter estimate in the residual stream, is rejected: the best probe reaches R^2=0.19, and steering and patching along it have no effect. Exploratory analyses locate the estimate instead in the attention key–value memory. Fast weights decode it (R^2=0.88 − −0.89, against $\leq$ 0.08 − −0.14 for the residual stream). The predictor's query Jacobian equals the Bayes estimate (R^2=0.92 − −0.92). Patching *only* context keys and values transfers 0.91 − −0.94 of the behaviour. The same holds for physical parameters in MuJoCo; on stick–slip friction the transformer beats physics-feature regression by 3.2×. A transformer controller, trained by imitation and then on closed-loop cost, learns to excite the plant with no identification objective (4.20 nats more information than exploration-free certainty equivalence) and matches classical adaptive controllers in median cost, but it fails on 10.0% of plants, almost all open-loop unstable, against 1.4% for dithered certainty equivalence. Code, pre-registration, deviation log and all numbers are public.

Keywords: in-context learning, adaptive control, system identification, dual control, mechanistic interpretability

1. Introduction

A robot picking up an unfamiliar object, or a legged robot stepping onto ice, must adapt to physics it does not know from a few seconds of interaction. A popular recipe trains a sequence model across randomised physics and lets it adapt *in context*, from its own history, with no gradient steps at deployment (Kumar et al., 2021; Duan et al., 2016; Forgione et al., 2023; Busetto et al., 2024). These models work, but it is not clear what they compute. Do they identify the plant and then plan? Do they implement a classical estimator? Do they explore?

For linear-Gaussian dynamics these questions have exact answers, so they can be tested. Transformers trained on random linear systems are known to *behave* like the Kalman filter or least squares (Balim et al., 2023; Akram and Vikalo, 2026). They are also known to be *able to represent* such estimators (Goel and Bartlett, 2024; Cole et al., 2025; Vlădărean et al., 2026). What has not been measured is what is inside a trained in-context system-identifier, or what an in-context *controller* does about the dual effect. The second question matters because, under exploration-free feedback, closed-loop

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data do not identify the plant (Feldbaum, 1960; Åström and Wittenmark, 1995; Ljung, 1999). Akram and Vikalo (2026) list the first question as future work.

We study both questions in a pre-registered design. Hypotheses, metrics, seeds and tests were committed before the main experiments, and deviations are logged. The design combines exact optimal references, causal interventions and three settings of increasing difficulty: linear prediction, MuJoCo contact physics (Todorov et al., 2012), and adaptive control. Our contributions:

- **Theory (§3).** An exact Bayes-risk identity for in-context prediction under any non-anticipating policy, with a parameter-free context-length law $t\,\mathcal{R}_t^\star \to np\sigma^2$. A closed-loop non-identifiability identity: the information gained along unexcited directions equals the excitation energy, and is exactly zero under pure feedback. A two-layer construction that gathers the exact sufficient statistics, and a first-order account of how a near-uniform attention head can apply a parameter estimate that is stored only in its key–value memory (Remark 5).

- **What the trained predictor computes (§5).** It is Bayes-optimal while data are scarce, and then it settles at an approximation floor. Our pre-registered mechanism, a linearly readable and causally used parameter estimate in the residual stream, is *rejected*: the probe reaches $R^2{=}0.19$, and steering and subspace patching have no effect. Exploratory analyses show instead that the estimate lives in the *key–value memory* of the last two blocks. It is readable from the fast weights ($R^2{=}0.88--0.89$), it equals the predictor's query Jacobian ($R^2{=}0.92--0.92$ against the Bayes estimate), and patching only context keys and values transfers $0.91--0.94$ of the behaviour.

- **Contact physics (§6).** On a pushed block with Coulomb friction the transformer beats physics-feature RLS by $3.2\times$, and on a pendulum it beats RLS even though RLS has the true model class. Physical parameters follow the same pattern: they are weakly readable from activations ($1/m$: $R^2{=}0.32--0.44$), but strongly readable from the key–value memory ($0.97--0.98$) and from the model's input sensitivity ($r{=}0.99--0.99$).

- **In-context control (§7).** A transformer controller trained by imitation and then on cost discovers excitation (4.20 nats of information above exploration-free certainty equivalence, of which 2.19 come from cost training), and it matches dithered certainty equivalence and Thompson sampling in median cost. Its failure rate, however, is 10.0% against 1.4%, concentrated on open-loop-unstable plants, and cost fine-tuning makes those failures more severe. The controller stores its estimate of the actuation matrix in the same key–value memory.

2. Related work

In-context learning of function classes. Transformers trained on random regression tasks match least squares (Garg et al., 2022), interpolate between ridge and OLS with intermediate quantities decodable by probes in static regression (Akyürek et al., 2023), can implement GD (von Oswald et al., 2023; Ahn et al., 2023; Zhang et al., 2024), higher-order iterations (Fu et al., 2023) and algorithm selection (Bai et al., 2023), and approximate posterior predictives (Müller et al., 2022; Raventós et al., 2023). We move these questions to closed-loop dynamical data, where the regressors depend on the past and on the policy, and where the dual effect has no i.i.d. analogue.

In-context learning of dynamical systems. Meta-trained transformers predict, filter and control unseen systems (Forgione et al., 2023; Piga et al., 2023; Busetto et al., 2023; Balim et al., 2023; Akram and Vikalo, 2026; Busetto et al., 2024). Constructions and approximation theory exist for Kalman filtering (Goel and Bartlett, 2024), least-squares learning of LDS (Cole et al., 2025) and single attention layers (Vlădărean et al., 2026). All of these are behavioural or representational; none intervenes on a trained model.

Adaptation from history, in-context RL, fast weights. RMA (Kumar et al., 2021) regresses a history encoding onto privileged physical parameters; probing such latents gives mixed results (Chamachot, 2026; Zhang et al., 2026). Meta-RL agents can learn to explore (Duan et al., 2016; Zintgraf et al., 2020), algorithm distillation imitates learning histories (Laskin et al., 2023), and in-context RL algorithms are representable in tabular MDPs (Liang and Lai, 2026). In adaptive LQR, certainty equivalence with dithering is rate-optimal (Mania et al., 2019; Simchowitz and Foster, 2020); these methods are our baselines. We use linear probes (Alain and Bengio, 2016) and activation patching (Zhang and Nanda, 2024; Heimersheim and Nanda, 2024), always paired with interventions, and the view of attention as reading and writing fast weights (Ba et al., 2016; Schlag et al., 2021; Dai et al., 2023) motivates our key–value analysis.

3. Theory

Setting. A plant $x_{s+1} = \theta z_s + w_s$ with regressor $z_s = (x_s, u_s) \in \mathbb{R}^p$, unknown $\theta = [A\ B] \in \mathbb{R}^{n\times p}$, and noise $w_s \sim \mathcal{N}(0, \sigma^2 I_n)$ i.i.d. and independent of everything else. The rows θ_i of θ have independent priors $\mathcal{N}(0, \Lambda_0^{-1})$. Inputs are chosen by an arbitrary *non-anticipating* policy: u_s is a measurable function of the past data $\mathcal{D}_s = (x_0, u_0, \dots, x_s)$ and of independent randomness, so it does not depend on θ except through $\mathcal{D}_s$. An *in-context predictor* is any map $\hat{y}_t = f_t(\mathcal{D}_t, u_t)$, of which a causal transformer is one example. Write $S_t = \sum_{s<t} z_s z_s^\top$ and $C_t = \sum_{s<t} z_s x_{s+1}^\top$.

Theorem 1 (Exact Bayes risk of in-context prediction) *Under the untruncated Gaussian prior, for every in-context predictor and every t,*

$$\mathbb{E}\,\|\hat{y}_t - \theta z_t\|^2 \;\geq\; \mathcal{R}_t^\star := n\sigma^2\,\mathbb{E}\big[z_t^\top(\sigma^2\Lambda_0 + S_t)^{-1} z_t\big], \tag{1}$$

with equality iff $\hat{y}_t = C_t^\top(\sigma^2\Lambda_0 + S_t)^{-1} z_t$ almost surely. This is ridge regression, i.e. RLS initialised at the prior.

Proof The likelihood of $\mathcal{D}_t$ factorises into the policy factors $p(u_s \mid \mathcal{D}_s)$, which do not depend on θ, and $\prod_{s<t}\mathcal{N}(x_{s+1}; \theta z_s, \sigma^2 I)$. Hence the posterior of each row is Gaussian, $\theta_i \mid \mathcal{D}_t \sim \mathcal{N}(m_i, \Sigma_t)$, the rows are independent, and $\Sigma_t = (\Lambda_0 + \sigma^{-2}S_t)^{-1} = \sigma^2(\sigma^2\Lambda_0 + S_t)^{-1}$ and $[m_1, \dots, m_n] = \Sigma_t\,\sigma^{-2}C_t$. Since u_t is conditionally independent of θ given $\mathcal{D}_t$, the posterior given $(\mathcal{D}_t, u_t)$ is the same. For any $(\mathcal{D}_t, u_t)$-measurable $\hat{y}_t$, the bias–variance identity gives $\mathbb{E}[\|\hat{y}_t - \theta z_t\|^2 \mid \mathcal{D}_t, u_t] = \|\hat{y}_t - m z_t\|^2 + \sum_i z_t^\top \Sigma_t z_t$. Taking expectations yields (1), with equality iff $\hat{y}_t = m z_t$. ∎

Corollary 2 (A parameter-free context-length law) *Fix θ and suppose that, conditionally on θ: (a) $S_t/t \to \Sigma_\theta \succ 0$ almost surely; (b) $\mathbb{E}[z_t z_t^\top \mid \theta] \to \Sigma_\theta$; (c) the families $\{t\, z_t^\top(\sigma^2\Lambda_0 + S_t)^{-1} z_t\}_t$ and $\{\|z_t\|^2\}_t$ are uniformly integrable. Then the conditional Bayes risk satisfies $t\,\mathcal{R}_t^\star(\theta) \to n\,p\,\sigma^2$. If, in addition, $\sup_t t\,\mathcal{R}_t^\star(\theta)$ is integrable over the prior, then $t\,\mathcal{R}_t^\star \to np\sigma^2$. The limit does not*

depend on the prior, on θ, or on the input distribution, because $\mathrm{tr}(\Sigma_\theta^{-1}\Sigma_\theta) = p$ *whatever* Σ_θ *is. Conditions (a)–(b) are standard for stable plants driven by i.i.d. inputs (as in Experiment A); we check the limit numerically rather than verify (c).*

Proof [Proof sketch] $t(\sigma^2\Lambda_0 + S_t)^{-1} \to \Sigma_\theta^{-1}$ a.s.; uniform integrability exchanges limit and expectation; and $\mathrm{tr}(\Sigma_\theta^{-1}\Sigma_\theta) = p$. Full proof in Appendix A. ■

Theorem 1 turns "is the transformer Bayes-optimal?" into a measurable gap. Corollary 2 gives a target curve $\mathcal{R}_t^\star \approx np\sigma^2/t$ that involves no fitted constants. (In Experiment A, the prior is truncated to stable A. The exact optimum is then the truncated posterior mean, which we compute by Monte Carlo. The truncation only matters at short contexts.)

Proposition 3 (Closed-loop non-identifiability; the dual effect) *Let the policy be* $u_s = Kx_s + \varepsilon_s$ *for a fixed* $K \in \mathbb{R}^{m\times n}$ *and an excitation sequence* ε_s*. For any* $b \in \mathbb{R}^m$*, let* $v_b = (-K^\top b,\ b) \in \mathbb{R}^p$*. Then for all* t*,*

$$v_b^\top \Sigma_t^{-1} v_b \ = \ v_b^\top \Lambda_0 v_b + \sigma^{-2} \textstyle\sum_{s<t} (b^\top \varepsilon_s)^2. \tag{2}$$

In particular, without excitation ($\varepsilon \equiv 0$), the posterior precision of every row along v_b stays at its prior value for every context length. By Cauchy–Schwarz, the posterior variance satisfies $v_b^\top \Sigma_t v_b \geq \|v_b\|^4/(v_b^\top \Lambda_0 v_b)$ for every t. This bound does not depend on t, and it equals the prior variance when Λ_0 is isotropic. So the combination $\theta v_b = (B - AK^\top)b$ is never learned, and no in-context learner can predict along it better than a constant above the noise floor.

Proof $v_b^\top z_s = -b^\top K x_s + b^\top(Kx_s + \varepsilon_s) = b^\top \varepsilon_s$, so $v_b^\top S_t v_b = \sum_{s<t}(b^\top \varepsilon_s)^2$. Substitute into $\Sigma_t^{-1} = \Lambda_0 + \sigma^{-2} S_t$. The variance bound follows from $(v^\top v)^2 \leq (v^\top \Sigma v)(v^\top \Sigma^{-1} v)$. ■

The extreme case is $K = 0$ with a prior mean of zero for B. A certainty-equivalent controller then outputs $u \equiv 0$ forever and never learns anything about B; Experiment C shows this failure exactly. A good adaptive controller must therefore *spend* control effort on excitation. The information it gathers along the unexcited directions equals the energy of that excitation, which we measure as $\log\det(\Lambda_0 + \sigma^{-2} S_8)$.

Proposition 4 (Two attention layers suffice to gather the sufficient statistics) *Let token* t *be* $e_t = (z_t, 1, \pi_t)$*, with* $\pi_t \in \{0,1\}^T$ *a one-hot position. There is a two-layer causal* linear-*attention network (one head in layer 1,* p *heads in layer 2, no MLPs) whose residual stream at every position* t *contains* S_t *and* C_t exactly*, together with* z_t *and* t*. Consequently, the Bayes predictor of Theorem 1 is a fixed continuous function* $\phi(S_t, C_t, z_t)$ *of the token-*t *residual alone. A token-wise MLP approximates it uniformly on any compact set, so in this construction all further depth is free for the "solve"* ϕ *rather than for gathering data.*

Proof [Proof sketch] A previous-token head pairs z_{s-1} with x_s; then p linear-attention heads with constant queries accumulate the columns of $[S_t;\ C_t^\top]$. Appendix A gives the weights. ■

Remark 5 (Where an estimate can live without appearing in any activation) *Consider a softmax head with keys* k_s*, values* v_s *and query* q*, and suppose its scores are small,* $|q^\top k_s|/\sqrt{d_h} \ll 1$*.*

This is the near-uniform regime, with normalised attention entropy close to 1. Expanding the softmax to first order,

$$\sum_s \mathrm{softmax}_s\, v_s \;\approx\; \bar{v} + \tfrac{1}{\sqrt{d_h}}\,\widehat{\mathrm{Cov}}(v,k)\,q, \qquad \widehat{\mathrm{Cov}}(v,k) = \tfrac{1}{t+1}\textstyle\sum_{s\le t}(v_s-\bar{v})k_s^\top. \tag{3}$$

The head is therefore an affine map of the query, and its matrix is the key–value covariance ("fast weights" (von Oswald et al., 2023)), which is spread over all context positions. A transformer can thus apply a parameter estimate to the query, with the predictor's Jacobian $\partial\hat{y}_t/\partial z_t$ playing the role of $\hat{\theta}_t$, while $\hat{\theta}_t$ never appears as a vector in the residual stream. Two predictions follow. A residual probe for $\hat{\theta}$ can fail even when the model is near Bayes-optimal; a probe on $\widehat{\mathrm{Cov}}(v,k)$, and patching of keys and values alone, should succeed. Section 5 confirms both.

Proposition 4 is in the spirit of earlier constructions (Akyürek et al., 2023; Bai et al., 2023; von Oswald et al., 2023), and we do not claim it as a contribution in itself. Its role is to make a *falsifiable* mechanistic prediction for trained models. If a trained transformer works this way, a linear probe should decode the normalised statistics $(S_t, C_t)/(t{+}1)$ no later in depth than it decodes the posterior mean $\hat{\theta}_t$ (H2c). If it instead refines $\hat{\theta}$ iteratively in the style of GD (von Oswald et al., 2023), $\hat{\theta}$ should become decodable gradually, with no early peak for the statistics.

4. Experimental design

Pre-registration. Before the main experiments we committed the hypotheses, metrics, seeds, thresholds and analysis plan (`PREREGISTRATION.md`, git commit `d134c5a`). Deviations are listed in `docs/DEVIATIONS.md` and in Appendix B. All confirmatory claims use a hierarchical bootstrap (training seeds, then test episodes; 10^4 resamples) with Holm–Bonferroni correction across the 13 confirmatory tests.

Model. Every experiment uses the same GPT-style causal decoder: pre-LN, $d{=}128$, 4 layers, 4 heads, MLP width 512, learned positions, about 0.8M parameters. Each model is trained from scratch with AdamW and three seeds, on a 4 GB laptop GPU. "Residual stream ℓ" means the input to block $\ell{+}1$ ($\ell{=}0$: embeddings).

A: in-context prediction of linear systems. $x \in \mathbb{R}^2$, $u \in \mathbb{R}$ ($p{=}3$), $\sigma{=}0.1$. The prior is $A_{ij} \sim \mathcal{N}(0, 0.4^2)$, $B_i \sim \mathcal{N}(0, 0.5^2)$, truncated to $\rho(A) < 0.95$, with $u_t \sim \mathcal{N}(0,1)$ and $T{=}48$. Token t is (x_t, u_t) and the target is x_{t+1}. Batches are freshly sampled for 20k steps. References: the *exact* Bayes predictor (truncated posterior mean, Monte Carlo with 2048 samples), ridge/RLS, OLS, one-step GD with tuned step size (von Oswald et al., 2023), and zero. Excess risk is measured against the noiseless mean θz_t.

B: MuJoCo physics. *Pendulum*: point mass on a 0.5 m rod, with $m \sim U[0.5, 2]$, joint damping $c \sim U[0.02, 0.3]$, $g \sim U[4, 16]$, random torques. Feature regression on $(u, \dot{q}, \sin q, 1)$ contains the true model class. *Pushed block*: a box on a plane with Coulomb contact friction $\mu \sim U[0.1, 1]$, $m \sim U[0.5, 3]$, and random forces $F \sim U[-15, 15]$ N. About half of the steps are stuck, so the stick–slip dynamics fall outside any linear-in-parameters model. Both systems use control step 0.05 s, unobserved process noise, 60k training episodes and $T{=}48$. Baselines: simulation with the true parameters, simulation with nominal parameters, physics-feature RLS, and a no-context MLP.

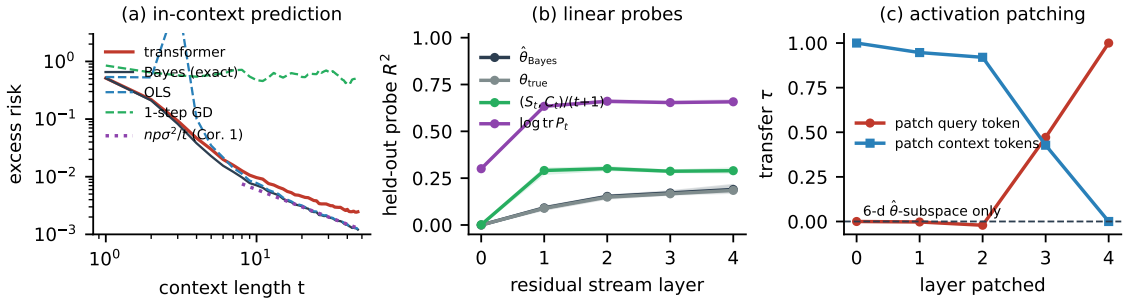


Figure 1: **A, pre-registered analyses** (3 seeds). (a) Excess risk vs. context length. The transformer tracks the exact Bayes predictor while data are scarce, then flattens at an approximation floor. The Bayes curve follows the parameter-free law of Cor. 2. (b) Held-out linear probes of the residual stream (band: range over seeds). The estimate $\hat{\theta}_t$ is barely decodable; the posterior uncertainty is. (c) Activation patching of episode A into B. Patching the query token transfers A's behaviour only from layer 3 on. Patching the 6-d $\hat{\theta}$-probe subspace transfers nothing (dashed).

C: in-context adaptive control. $A = 0.9I + \mathcal{N}(0, 0.25^2)$ truncated to $\rho(A) < 1.1$, so some plants are open-loop unstable. $B = r(\cos\phi, \sin\phi)$ with ϕ uniform, so the sign and direction of actuation are unknown. Cost $J = \sum_{t<48} \|x_t\|^2 + 0.1u_t^2 + \|x_{48}\|^2$, $|u| \leq 5$. The transformer reads (x_t, u_{t-1}), outputs u_t, and is trained by back-propagating J through the differentiable plant. It never sees θ and has no system-ID loss. Baselines, all tuned on separate plants: oracle finite-horizon LQR; the best constant gain over the prior ("fixed"); certainty equivalence (CE: ridge estimate, then LQR on the remaining horizon); CE with decaying dither $s_0/\sqrt{t+1}$ (Simchowitz and Foster, 2020); and explore-then-CE (system ID followed by control). Test plants and noise are shared across controllers.

5. Experiment A: what the in-context predictor computes

Behaviour: Bayes-optimal while data are scarce, then an approximation floor. The trained predictor is much closer to the exact Bayes predictor than to OLS or to one GD step. The mean algorithmic distance to OLS exceeds the distance to Bayes by 0.271 [0.181, 0.379], and the distance to GD exceeds it by 0.552 [0.496, 0.616] (H1a, H1b; Holm $p = 0.001$, $= 0.001$). For $t \leq 4$ its excess risk is within a factor 1.23 of Bayes. After that it decays more slowly than $1/t$ and settles about 1.3×10^{-3} above Bayes. Meanwhile $t\,\mathcal{R}_t^\star$ reaches 0.057 at t=47: Theorem 1 gives 0.058 and Corollary 2 predicts $np\sigma^2$=0.06. The pre-registered tolerance H1c is **rejected**: the mean excess ratio is 1.59 [1.56, 1.62], against a threshold of 1.25. This is the dynamical analogue of the observation that trained transformers fall short of least-squares rates (Hill et al., 2025; Cole et al., 2025). Section 8 tests whether longer training or larger models close the gap.

The pre-registered mechanism is rejected: no estimate in the residual stream. The best linear probe of the residual stream for $\hat{\theta}_t$ reaches only R^2=0.19 [0.17, 0.21] (H2a, threshold 0.8: rejected). It beats the probe for the true θ by just 0.005 (H2b: significant but negligible). The normalised sufficient statistics are partly decodable ($R^2 \leq 0.30$) and peak at layers 1–2, before $\hat{\theta}$ peaks at layer 4,

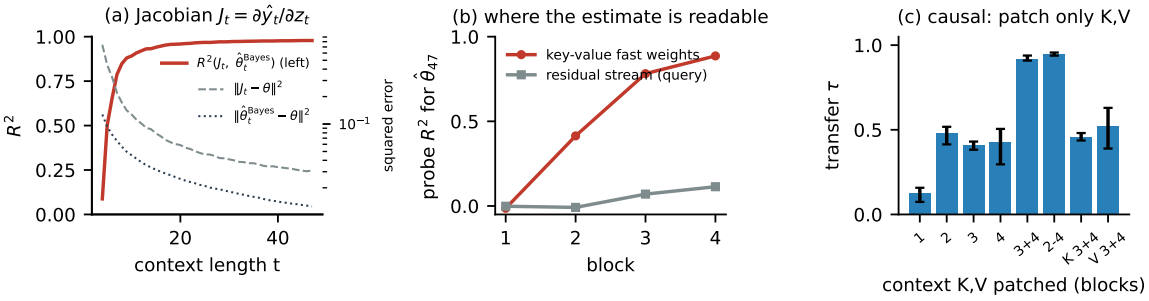


Figure 2: **A, exploratory follow-up: where the estimate lives** (3 seeds). (a) The predictor's Jacobian with respect to its query, $J_t{=}\partial\hat{y}_t/\partial z_t$, matches the Bayes estimate. Its error to the truth (dashed) tracks the Bayes estimate's error (dotted). (b) $\hat{\theta}_{47}$ is linearly readable from the per-block key–value fast weights $\widehat{\mathrm{Cov}}(v,k)$ but not from the residual stream at the same position. (c) Patching *only* the context keys and values of blocks 3–4, leaving every activation of the query untouched, transfers over 90% of the other episode's behaviour. Keys or values alone transfer about half.

in every seed. So H2c holds formally but is uninformative. What *is* linearly decodable is the model's uncertainty: $\log \mathrm{tr}\, P_t$ reaches $R^2{=}0.66$. Steering along the $\hat{\theta}$-probe directions does not move the prediction (slope -0.0000; H3a rejected). Swapping the 6-d probe subspace of the query between episodes transfers $\tau{=}0.0001$ of the behaviour, which is less than a random 6-d subspace (0.067; H3b rejected). Whole-stream patching (Fig. 1c) shows that parameter information enters the query position only in blocks 3–4: $\tau_{\text{last}}{=}0.38 - -0.62$ after block 3, while patching all context positions at layer 2 still transfers ≥ 0.91.

Exploratory: the estimate lives in the keys and values and acts through the query Jacobian. These analyses were added after the pre-registered ones failed; Remark 5 says where to look. *The answer is linearly represented, the parameter is not:* a linear probe recovers the Bayes *prediction* $\hat{\theta}_t z_t$ with $R^2{=}0.88 - -0.92$ after block 3 and 0.97–0.98 after block 4, while an MLP probe recovers $\hat{\theta}_t$ itself only to 0.72. *The Jacobian is the estimate:* the predictor is near-linear in its query (counterfactual queries are explained with $R^2{=}0.87 - -0.96$ by its Jacobian, slope 0.97–0.99 against the Bayes map), and $J_t{=}\partial\hat{y}_t/\partial z_t$ matches $\hat{\theta}_t^{\text{Bayes}}$ with $R^2{=}0.92 - -0.92$, more closely than the truth (0.89–0.89) or OLS (0.90–0.90) (Fig. 2a). *It is stored in key–value memory:* the per-head fast weights $\widehat{\mathrm{Cov}}(v,k)$ of block 4 predict $\hat{\theta}_{47}$ with $R^2{=}0.88 - -0.89$ (block 3: 0.78–0.79), while the residual stream at the same position gives at most 0.08–0.14 (Fig. 2b). *The memory is causal:* replacing only the context keys and values of blocks 3–4 by another episode's, with the query's residual stream, its attention queries and all MLPs untouched, transfers $\tau{=}0.91 - -0.94$ of that episode's behaviour (blocks 2–4: 0.94–0.96; keys alone 0.44–0.48, values alone 0.39–0.63; Fig. 2c), carried by 2–3 near-uniform heads per block, as in Remark 5. *Probe directions are not the model's directions:* even for the well-decoded prediction, steering along the probe has slope 0.08 and subspace patching transfers 0.08. In short, the transformer implements near-Bayes system identification without writing $\hat{\theta}$ into any activation: the estimate is spread over the key–value memory of the context and is applied to the query by attention. This is why RMA-style "latent parameter" probes of activations (Kumar et al., 2021; Chamachot, 2026) can fail even when adaptation works.

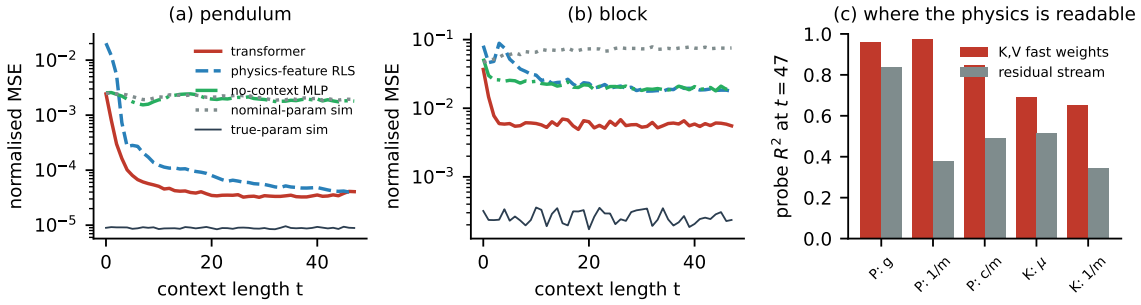


Figure 3: **B** (3 seeds). (a, b) Normalised one-step error vs. context length. On the pendulum the transformer beats physics-feature RLS, even though RLS has the true model class up to discretisation. On the stick–slip block it beats RLS by 3.2×. (c) Probe R^2 at t=47 for each physical parameter, from the key–value fast weights (best block) vs. from the residual stream (P: pendulum, K: block, B: bouncing ball).

6. Experiment B: nonlinear and contact physics in MuJoCo

Prediction. Averaged over t=24..47, the normalised error on the pushed block with Coulomb friction is 0.0058 for the transformer and 0.019 for physics-feature RLS (H4a: difference 0.0131 [0.0128, 0.0134], Holm $p = 0.001$). For comparison, the no-context MLP gets 0.019, nominal-parameter simulation 0.075, and simulation with the true parameters 0.00026. On the pendulum, where feature RLS has the right model class up to discretisation, the transformer (3.5e-05) still beats it (5e-05). Nominal simulation (0.002) and the no-context MLP (0.0019) are $\approx 58\times$ worse, so the gain comes from in-context adaptation.

Mechanism. The pre-registered probes again fail for the parameters that enter the dynamics multiplicatively. Residual-stream R^2 at t=47 is 0.82–0.86 for g, 0.32–0.44 for $1/m$ and 0.51–0.52 for μ (H4c rejected). Steering along the $1/m$ probe moves the input sensitivity the right way in only 42% (pendulum) and 71% (block) of episodes (H4d rejected). The exploratory analyses of §5 transfer. The model's own sensitivity $\partial \dot{q}_{t+1}/\partial u_t$, its functional estimate of $1/m$, correlates with the true $1/m$ at r=0.99 − −0.99 on the pendulum; on the block it is r=0.42 − −0.43, because the sensitivity is zero while stuck. The key–value fast weights decode $1/m$ with R^2=0.97 − −0.98 (residual: 0.32–0.44), and on the block they decode μ at 0.68–0.72 (residual 0.51–0.52). **[bouncer pending]**

7. Experiment C: does an in-context controller explore?

The dual effect in the baselines. With the prior mean B=0, certainty equivalence (CE) computes a zero gain, outputs $u \equiv 0$, and so never learns B. Its information after 8 steps stays at the prior level (10.0 nats, against 9.9 for the oracle, which needs none), it fails on 30.2% of plants, and its median cost is 2.81. A constant gain cannot help when the actuation direction is unknown (median 2.80). Any excitation fixes this: decaying dither (median 1.57, 1.4% failures, 15.6 nats), explore-then-commit (1.76, 3.4%, 16.1) and Thompson sampling (1.78, 1.4%, 16.4).

The transformer explores, without being told to. Pre-registered pure BPTT did not train (Appendix, Table 1); the controller is a DAgger warm start followed by cost fine-tuning. After the warm

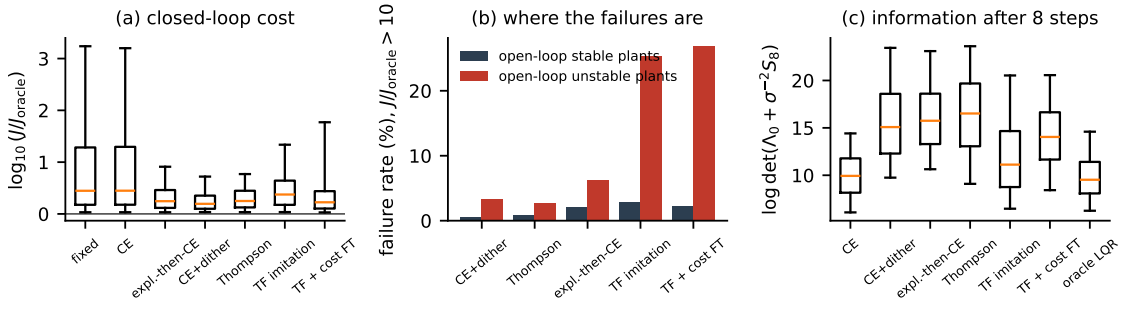


Figure 4: **C** (4096 paired test plants, 3 seeds). (a) Normalised cost J/J_{oracle}; boxes span the 5–95% range. Certainty equivalence without excitation never moves u, so it learns nothing (Prop. 3). (b) Failure rate on open-loop stable vs. unstable plants. (c) Information gathered in the first 8 steps.

start it gathers 12.0 nats; cost fine-tuning adds 2.19 [2.06, 2.37] more, to 14.2. That is 4.20 [4.06, 4.31] above CE (H5d, Holm $p = 0.001$). So a controller with no identification objective learns to excite the plant, and most of that excitation is learned from the cost, not copied from the LQR teacher. It still excites less than dither (-1.41 [-1.55, -1.29]) and Thompson sampling (-2.18 [-2.32, -2.05]).

Cost: competitive in the typical case, unsafe in the tail. The median normalised cost is 1.69 (seeds 1.53–1.86), against 1.57 for dither and 1.78 for Thompson sampling; per plant, the transformer wins against Thompson sampling 51% of the time and against dither 43% (paired median differences 0.01 [-0.09, 0.11] and -0.10 [-0.22, -0.00]). But it fails on 10.0% of plants against 1.4%. The failures sit almost entirely on the 31% of plants that are open-loop unstable: 27% there against 2.3% on stable plants (dither: 3.3% and 0.5%), Fig. 4b. Controllability is not the issue. Because the mean is dominated by these failures, the transformer beats the constant gain (H5a, Holm $p = 0.015$) but *not* CE on the pre-registered mean (H5b: 111.60 [-149.21, 343.32], $p = 1.000$), and it is far worse than every excited baseline in the mean (vs. dither: -379.87 [-648.12, -152.71]).

What cost fine-tuning buys, and what it costs. Relative to the warm start, fine-tuning on $\log(J/J_{\text{oracle}})$ improves the median (2.38 to 1.69) and the excitation (above), leaves the failure rate unchanged (9.9% to 10.0%), and makes the failures far more severe: the mean rises from 22.69 to 382.19 (difference -359.50 [-626.51, -130.78]). The log objective compresses the tail, so the optimiser trades rare catastrophes for typical-case gains. A raw cost would weight those cases, but it did not train (Table 1). The safe in-context controller on unstable plants remains an open problem.

Exploratory: the controller identifies the plant in its key–value memory. On its own closed-loop trajectories, a residual-stream probe recovers the actuation matrix B with R^2=0.47 (pooled over $t \geq 4$), and flipping the probed estimate flips the action in only 0% of cases, as in §5. **[mechC pending]**

8. Scale and training length

Table 2 (appendix) asks whether the approximation floor of §5 is an artefact of a small model or short training, and whether the Jacobian account survives a larger system. **[scaleprose pending]**

9. Discussion and limitations

What the transformer computes. In all three settings the trained models are good in-context system identifiers. They match the exact Bayes predictor at short contexts, beat physics-feature regression on contact dynamics, and adapt within a few steps. Their *form* of identification is not the one we pre-registered. There is no linearly readable, causally used parameter vector in the residual stream. The estimate sits in the key–value memory of the last two blocks, and attention applies it to the query, so it shows up as the input–output Jacobian of the model. Remark 5 explains why this is a natural solution for near-uniform heads, and Proposition 4 explains why two attention layers suffice to gather the required statistics. For interpretability of adaptive controllers (and of RMA-style policies), the lesson is to probe the *function* (Jacobians) and the *memory* (keys and values), not only the activations.

Why pre-registration mattered. Without it, a residual probe with $R^2{=}0.19$ and a significant but negligible gain over the truth (H2b) could have been written up as "the model represents its estimate". The pre-registered thresholds and the causal tests H3a/H3b rejected that story; the alternative was found with exploratory tools, is labelled as such, is tested causally (Fig. 2c), and is tested out-of-sample on models trained after the hypothesis was formed (the bouncing ball, the scale study).

Limitations. (i) *Scale.* The models have 0.2–3M parameters, the linear systems have at most 24 unknowns, and the MuJoCo systems have one degree of freedom. Whether the key–value account holds for large robot foundation models is open. (ii) *Post-hoc mechanism.* The fast-weight account was formulated after the pre-registered probes failed. Its main evidence comes from the same models, with out-of-sample support from new models only. (iii) *First-order picture.* Remark 5 linearises softmax attention. The heads are near-uniform but not exactly, and the account explains roughly 90% of the transferable behaviour, not all of it. (iv) *Approximation floor.* The predictors stay a factor 1.59 above the Bayes risk, averaged over contexts (H1c rejected). **[scaleclaim pending]** (v) *Control recipe.* Pure closed-loop BPTT, as pre-registered, did not train reliably on open-loop-unstable plants. Our controller conclusions are about a DAgger warm start followed by cost fine-tuning, and other recipes (e.g. RL) may behave differently. (vi) *Authorship.* This is single-author independent work produced with substantial AI assistance (see below). Every number is regenerated by the public scripts, and the pre-registration and deviation log are public.

Generative AI use

Generative AI tools (Claude) were used for literature search and synthesis, code implementation, mathematical derivations, and writing. The author reviewed the content, every reported number is regenerated by the scripts in the public repository, and the author takes responsibility for the final content.

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Appendix A. Omitted proofs

Proof of Corollary 2. Let $M_t = t(\sigma^2\Lambda_0 + S_t)^{-1}$, so $M_t \to \Sigma_\theta^{-1}$ a.s. by (a). Then $|z_t^\top(M_t - \Sigma_\theta^{-1})z_t| \le \|z_t\|^2\|M_t - \Sigma_\theta^{-1}\|$, which tends to 0 in probability because $\|z_t\|^2$ is tight by (c). The difference is uniformly integrable as a difference of uniformly integrable families, so its expectation vanishes. Hence $\mathbb{E}[z_t^\top M_t z_t \mid \theta] = \mathrm{tr}(\Sigma_\theta^{-1}\mathbb{E}[z_t z_t^\top \mid \theta]) + o(1) \to p$ by (b). Multiply by $n\sigma^2$. The unconditional statement follows by dominated convergence.

Proof of Proposition 4. *Layer 1 (previous-token head).* Set $q_t = \Pi\pi_t = \pi_{t-1}$, with Π the downshift and $\pi_{-1} = 0$; set $k_s = \pi_s$ and $v_s = z_s$. Linear attention outputs $\sum_{s\le t} v_s\, k_s^\top q_t = v_{t-1}$. The residual becomes $(z_t, 1, \pi_t, z_{t-1})$, with $z_{-1} = 0$. *Layer 2 (moment heads).* Head $j \in \{1..p\}$ uses the constant query $q_t = 1$, the key $k_s = (z_{s-1})_j$, and the value $v_s = (z_{s-1}, x_s)$. It outputs $\sum_{s\le t}(z_{s-1}, x_s)(z_{s-1})_j$, which is the j-th column of $[S_t; C_t^\top]$, because the pairs (z_{s-1}, x_s), $1 \le s \le t$, are exactly the context pairs. Writing the p head outputs to disjoint residual coordinates gives S_t and C_t. The map ϕ is continuous on $\{\sigma^2\Lambda_0 + S \succ 0\}$, so the universal approximation theorem applies on compacta.

Appendix B. Deviations from the pre-registration

(1) We measure excess risk against the noiseless mean θz_t instead of as MSE $- n\sigma^2$. The two have the same expectation and ours has lower variance. (2) $\Lambda_0{=}I$ in the information measure, which the pre-registration did not fix. (3) Training details not fixed by the pre-registration: the constant-gain baseline uses Adam, 1000 steps and batch 1024; the no-context MLP has $2{\times}256$ units and 5000

steps. (4) The controller uses a KV-cached incremental forward to fit in 4 GB. It is numerically identical to the full forward. (5) With a prior mean $B{=}0$, pre-registered CE degenerates to $u \equiv 0$; we report it as specified. (6) On the pendulum the parameter effects are small (nominal-parameter simulation already has 0.5% normalised error); we kept the design unchanged. (7) All analyses labelled *exploratory* were added after the pre-registered probes failed: the query Jacobian, fast-weight probes, KV-only patching, answer probes, MLP probes, the bouncing-ball system, the Thompson-sampling baseline and the scale study. (8) *Controller training.* Pure closed-loop BPTT on the cost, as pre-registered, was unstable in pilots (Table 1). The final controller uses a DAgger warm start (imitating the oracle LQR action from its own history) followed by BPTT fine-tuning on $\log(J/J_{\text{oracle}})$, with 8-step truncation and learning rate 10^{-4}, for the pre-registered 6000 BPTT steps. (9) Execution only: BPTT is replayed as a CUDA graph; the constant-gain baselines are trained on the CPU. The full log is in `docs/DEVIATIONS.md`.

Appendix C. Implementation details

Exact Bayes predictor (A). Each row of θ has the Gaussian posterior of Theorem 1, and the rows share their covariance. We draw 2048 joint samples, discard those with $\rho(A) \geq 0.95$, and average. The acceptance rate is 0.94 at $t{=}0$ and 0.996 at $t{=}47$. **Probes.** Ridge regression on standardised residuals, with α chosen by 5-fold CV grouped by episode, fitted on 2048 test episodes and scored on the other 2048. **Steering.** Let D be the probe readout matrix. The minimal-norm residual change that moves readout coordinate k by δ while leaving the others unchanged is $\delta\,(D^{+})_k$. **Patching transfer** is $\tau = \langle \hat{y}_{\text{patch}} - \hat{y}_B, \hat{y}_{A'} - \hat{y}_B \rangle / \|\hat{y}_{A'} - \hat{y}_B\|^2$, where A' is episode A's context followed by episode B's query token. **Compute.** Everything runs on a laptop RTX 3050 Ti (4 GB) with a 20-thread CPU, in about **[hours pending]** hours end to end.

Table 1: Controller-training pilots (seed 99, 1024 held-out plants). Cost is J/J_{oracle}; "fail" means $J/J_{\text{oracle}}{>}10$. †Intended lr 10^{-4}, but a bug ran these pilots at $3{\cdot}10^{-4}$. The final runs use 10^{-4}.

variant	step	median	mean	fail
BPTT, J/J_{oracle}, lr $3{\cdot}10^{-4}$	1000	10.49	298	51%
	1999	84.89	1146	96%
BPTT, $\log(J/J_{\text{oracle}})$, lr $3{\cdot}10^{-4}$	1000	2.41	300	21%
	1999	11.67	976	52%
BPTT, log, squashed inputs†	1000	137.67	76387	98%
	1999	156.09	51593	98%
same, 8-step truncated BPTT	1000	4.35	6542	36%
DAgger warm start (3000) + fine-tune (1500)†	0	2.69	204	19%
	1000	2.09	1070	24%
	1499	1.99	250	16%

[appextra pending]

Table 2: Scale study (exploratory, one seed unless stated). Ratio: excess risk relative to the Bayes (ridge) predictor, mean over $t{=}4..47$ and at $t{=}47$. $R^2(J)$: agreement of the query Jacobian with the ridge estimate, mean over $t \in \{4, 8, 16, 32, 47\}$.

model	change	params	ratio ($t \geq 4$)	ratio ($t{=}47$)	$R^2(J)$
A (3 seeds)	reference, 20k steps	0.8M	1.59	2.04	0.92